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Inventor(s)	Kuribayashi; Yasushi

Optical scanning device

Abstract

An optical scanning device according to an embodiment includes a light source, a first optical sheet, a second optical sheet, a holder, and a polygon mirror. The light source emits light. The first optical sheet is disposed on an optical path of the light emitted from the light source. The second optical sheet is disposed on the optical path. The holder includes a pair of engaging sections with which both end portions in a bending direction of the first optical sheet are engaged and with which both end portions in a bending direction of the second optical sheet are engaged. The holder holds the first optical sheet and the second optical sheet. The polygon mirror reflects the light from the light source guided via the first optical sheet and the second optical sheet and rotates to scan a scan surface present in a reflecting direction of the light.

Inventors: **Kuribayashi; Yasushi (Sunto Shizuoka, JP)**

Applicant: **TOSHIBA TEC KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **92544823**

Assignee: **TOSHIBA TEC KABUSHIKI KAISHA (Tokyo, JP)**

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Primary Examiner: Caley; Michael H

Attorney, Agent or Firm: Amin, Turocy & Watson, LLP

Background/Summary

FIELD

(1) Embodiments described herein relate generally to an optical scanning device that scans a scan surface with light emitted from a light source.

BACKGROUND

(2) As an image forming apparatus such as a digital multifunction peripheral or a printer installed in a workplace, there is an apparatus including a plurality of color image forming units in order to form a color image. An optical scanning device that exposes and scans photoconductive drums of the color image forming units sometimes include quarter wavelength plates between color light sources and polygon mirrors in order to align deflection states of lights from the color light sources. As the quarter wavelength plates, there is a quarter wavelength plate obtained by sandwiching a double refraction film with glass plates to form a rigid body and bonding and fixing the rigid body to another optical element.

(3) If it is attempted to incorporate a relatively inexpensive sheet-type wavelength plates in the device in order to reduce costs of the device, it is difficult to stably fix the wavelength plate because, for example, the wavelength plate creases. Since the sheet-type wavelength plate easily bends and has difficulty in independently standing, the sheet-type wavelength plate needs to be bonded and fixed to a holder or the like and used. Therefore, an assembly manhour for the device increases and an adhesive adheres to undesired parts of the wavelength plate. It is likely that optical performance of the wavelength plate cannot be sufficiently exerted.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram of an image forming apparatus according to an embodiment;
(2) FIG. **2** is a plan view of an optical scanning device incorporated in the image forming apparatus illustrated in FIG. **1**;
(3) FIG. **3** is a perspective view of the optical scanning device illustrated in FIG. **2**;
(4) FIG. **4** is a schematic diagram illustrating a light source of the optical scanning device;
(5) FIG. **5** is a schematic diagram illustrating an optical system of the optical scanning device;
(6) FIG. **6** is a front view illustrating a diaphragm plate included in the optical system illustrated in FIG. **5**;
(7) FIG. **7** is a front view illustrating a wavelength plate included in the optical system;
(8) FIG. **8** is a side view of a structure in which the diaphragm plate illustrated in FIG. **6** and the wavelength plate illustrated in FIG. **7** are attached to a sheet holder, the structure being viewed from an arrow F**8** direction of FIG. **5**;
(9) FIG. **9** is a front view illustrating a wavelength plate in a first modification;
(10) FIG. **10** is a side view of a structure in which the wavelength plate illustrated in FIG. **9** is attached to the sheet holder;
(11) FIG. **11** is a front view illustrating a diaphragm plate in a second modification;
(12) FIG. **12** is a front view illustrating a wavelength plate in the second modification; and
(13) FIG. **13** is a side view of a structure in which the diaphragm plate illustrate in FIG. **11** and the wavelength plate illustrated in FIG. **12** are attached to the sheet holder.

DETAILED DESCRIPTION

(14) An optical scanning device according to an embodiment includes a light source, a first optical sheet, a second optical sheet, a holder, and a polygon mirror. The light source emits light. The first optical sheet is disposed on an optical path of the light emitted from the light source in a state in which the first optical sheet is bent by elastic deformation. The second optical sheet is disposed on the optical path, capable of transmitting the light, and bendable by elastic deformation. The holder includes a pair of engaging sections with which both end portions in a bending direction of the first optical sheet are engaged by a restoration force based on the elastic deformation of the first optical sheet and with which both end portions in a bending direction of the second optical sheet are engaged by a restoration force based on the elastic deformation of the second optical sheet in a state in which the second optical sheet is bent at a curvature different from a curvature of the first optical sheet and in a same direction as the bending direction of the first optical sheet. The holder holds the first optical sheet and the second optical sheet in a state in which the first optical sheet and the second optical sheet are superimposed via a gap. The polygon mirror reflects the light from the light source guided via the first optical sheet and the second optical sheet and rotates to scan a scan surface present in a reflecting direction of the light.

(15) Several embodiments are explained below with reference to the drawings. In the embodiments, the same components are denoted by the same reference numerals and signs and redundant explanation of the components is sometimes omitted. In the drawings referred to in the following explanation, scales of sections are sometimes changed as appropriate. In the drawings, components are illustrated to be simplified or omitted in order to facilitate understanding of explanation.

(16) As illustrated in FIG. **1**, an image forming apparatus **1** includes an image reading section **10** and an image forming section **20**.

(17) The image reading section **10** includes an auto document feeder **11**. The auto document feeder **11** conveys a plurality of originals inserted in a stacked state to an original table one by one. The image reading section **10** reads image information from the original conveyed to the original table by the auto document feeder **11** or an original placed on the original table by a user and converts the image information into image data.

(18) The image forming section **20** forms an image on paper based on the image data read from the original by the image reading section **10** or image data transmitted from external equipment such as a personal computer. The image forming section **20** includes a paper feeding cassette **21**, an optical scanning device **30**, a photoconductive drum **22**, a developing device **25**, a fixing device **23**, and a paper discharge tray **24**.
(19) The paper feeding cassette **21** stores a plurality of pieces of paper in a stacked state. The image forming section **20** picks up the paper from the paper feeding cassette **21** piece by piece with a pickup roller and conveys the paper to the photoconductive drum **22** with a plurality of conveying rollers.
(20) A surface **221** of the photoconductive drum **22** is an example of the scan surface described in the claims of this application. A direction parallel to a rotation axis of the photoconductive drum **22** in the scan surface (a direction orthogonal to the paper surface in FIG. **1**) is a main scanning direction. A direction orthogonal to the main scanning direction in the scan surface (a direction in which the surface **221** of the photoconductive drum **22** rotates) is a sub-scanning direction. In the following explanation, in order to facilitate understanding of the explanation, all directions equivalent to the direction of the rotation axis of the photoconductive drum **22** are referred to as main scanning direction and all directions equivalent to a rotating direction of the surface **221** of the photoconductive drum **22** are referred to as sub-scanning direction irrespective of whether the directions are in the scan surface.
(21) The developing device **25** develops, with a developer, an electrostatic latent image based on image data formed on the surface **221** of the photoconductive drum **22** by the optical scanning device **30** explained below. The image forming section **20** transfers a developer image formed on the surface **221** of the photoconductive drum **22** onto the paper conveyed from the paper feeding cassette **21**. The fixing device **23** heats the developer image transferred on the paper and fixes the developer image on the paper. The image forming section **20** discharges the paper having passed through the fixing device **23** and subjected to image formation to the paper discharge tray **24**.
(22) As illustrated in FIGS. **2** and **3**, the optical scanning device **30** includes an optical system **31**, a light source **32**, a polygon mirror **33**, a scanning lens **34**, a mirror **35**, and a housing **36** in which these components are housed and fixed. The optical system **31** changes characteristics of light emitted from the light source **32** and guided to the polygon mirror **33**. The optical system **31** is explained in detail below.
(23) As illustrated in FIG. **4**, the light source **32** is, for example, a semiconductor laser array having a structure in which a junction layer **321** emitting light is sandwiched by two reflection layers **322**. The junction layer **321** includes a plurality of (in this embodiment, three) light emitting regions side by side in one direction. The plurality of light emitting regions are an example of the plurality of light emitting sections described in the claims of this application. Polarization directions of lights emitted from the light emitting regions are parallel to the junction layer **321**. Therefore, if the junction layer **321** (that is, the light source **32**) in a state in which the plurality of light emitting regions are arranged in the main scanning direction according to a scanning pitch in the sub-scanning direction on the scan surface is fixed to the housing **36** while being slightly tilted with respect to the main scanning direction, the polarization direction of the lights emitted from the light emitting regions changes to a direction slightly inclined from the main scanning direction and generally conforming to the main scanning direction.
(24) The polygon mirror **33** is a rotating polygon mirror including a plurality of (in this embodiment, six) reflection surfaces on the outer circumferential surface thereof. The polygon mirror **33** is rotatably attached to the housing **36** at an angle at which a rotation axis of the polygon mirror **33** is parallel to the sub-scanning direction. The polygon mirror **33** reflects light having passed through the optical system **31** on the plurality of reflection surfaces while rotating and deflects the light reflected on the reflection surfaces to the main scanning direction.
(25) The scanning lens **34** extends in the main scanning direction and is fixed to the housing **36**.

The scanning lens **34** converges the light reflected on the reflection surfaces of the polygon mirror **33** in the sub-scanning direction. The mirror **35** extends in the main scanning direction and is fixed to the housing **36**. The mirror **35** reflects the light transmitted through the scanning lens **34** toward the surface **221** of the photoconductive drum **22**. A plurality of mirrors **35** may be used in combination.

(26) In FIGS. **2** and **3**, parts of lights passing through the optical scanning device **30** are illustrated as rays L**1** and L**2**. The rays L**1** are parts of lights emitted from the light source **32**, having passed through the optical system **31**, and deflected by the polygon mirror **33**. The rays L**2** are parts of the lights after being reflected on the mirror **35**.

(27) As illustrated in FIG. **5**, the optical system **31** includes a collimator lens **311**, a diaphragm plate **312**, a wavelength plate **314**, and a cylindrical lens **313**. The diaphragm plate **312** is an example of the first optical sheet described in the claims of this application. The wavelength plate **314** is an example of the second optical sheet described in the claims of this application. The members **311**, **312**, **313**, and **314** of the optical system **31** are disposed side by side to be separated from one another in an optical axis direction. An arrow X illustrated in FIG. **5** indicates the main scanning direction, an arrow Y illustrated in FIG. **5** indicates an optical axis direction of light emitted from the light source **32**, and an arrow Z illustrated in FIG. **5** indicates the sub-scanning direction. In the following explanation, the X direction, the Y direction, and the Z direction are used for explanation concerning directions.

(28) The housing **36** integrally includes a first lens holder **361** that holds the collimator lens **311**, a second lens holder **362** that holds the cylindrical lens **313**, and a sheet holder **363** that holds the diaphragm plate **312** and the wavelength plate **314** to be superimposed. The sheet holder **363** is an example of the holding section described in the claims of this application.

(29) The collimator lens **311** transmits diverging light emitted from the light source **32** and converts the diverging light into parallel light. The first lens holder **361** fixes the collimator lens **311** to the housing **36** in a position where the collimator lens **311** is separated from and opposed to the light source **32** on an emission side of light emitted from the light source **32** (a downstream side in the Y direction). The first lens holder **361** holds the collimator lens **311** such that the center of the collimator lens **311** is located on the optical axis of the optical system **31**.

(30) As illustrated in FIG. **6**, the diaphragm plate **312** has a substantially rectangular plate shape and includes a substantially rectangular projecting section **3122** projecting from a lower end side **3121** in a direction (the first direction) orthogonal to the lower end side **3121**. Width W**1** in a direction (the second direction) orthogonal to the projecting direction of the projecting section **3122** is smaller than the width of the diaphragm plate **312**. The projecting section **3122** is an example of the first projecting section described in the claims of this application.

(31) The diaphragm plate **312** is a thin metal plate such as stainless steel that does not transmit light. The diaphragm plate **312** includes, near the center thereof, a rectangular opening section **3123** that allows light to pass. The diaphragm plate **312** blocks light other than the light passing through the opening section **3123** and defines the widths in the main scanning direction and the sub-scanning direction of the light passing through the opening section **3123**. The diaphragm plate **312** has a degree of thickness bendable by elastic deformation. In the following explanation, the length in an illustrated up-down direction of the diaphragm plate **312** excluding the projecting section **3122** is represented as H**1** and the width in an illustrated left-right direction of the projecting section **3122** is represented as W**1**. The length H**1** is length in a state in which the diaphragm plate **312** is not bent.

(32) As illustrated in FIG. **7**, the wavelength plate **314** has a substantially rectangular plate shape and includes a substantially rectangular projecting section **3142** projecting from a lower end side **3141** in a direction (the first direction) orthogonal to the lower end side **3141**. Width W**2** in a direction (the second direction) orthogonal to the projecting direction of the projecting section **3142** is smaller than the width of the wavelength plate **314**. The projecting section **3142** is an example of

the second projecting section described in the claims of this application.
(33) The wavelength plate **314** transmits light and changes a polarization direction of the light. If a wavelength plate **314** is a half wavelength plate, the wavelength plate **314** changes the polarization direction of the light by 45°. If the wavelength plate **314** is a quarter wavelength plate, the wavelength plate **314** changes linearly polarized light to circularly polarized light. The wavelength plate **314** has a structure in which a double refraction film is sandwiched by two PET (polyethylene terephthalate) sheets and has a degree of thickness (in this embodiment, approximately 0.1 mm) bendable by elastic deformation.
(34) The wavelength plate **314** has substantially the same shape as the shape of the diaphragm plate **312**. However, length H**2** in the illustrated up-down direction excluding the projecting section **3142** is slightly smaller than the length H**1** of the diaphragm plate **312**. The length H**2** is length in a state in which the wavelength plate **314** is not bent. Width W**2** of the projecting section **3142** of the wavelength plate **314** is the same as the width W**1** of the projecting section **3122** of the diaphragm plate **312**. The diaphragm plate **312** has higher rigidity and less easily bends than the wavelength plate **314**.
(35) By differentiating the length H**2** of the wavelength plate **314** from the length H**1** of the diaphragm plate **312**, it is possible to differentiate a curvature of the wavelength plate **314** from a curvature of the diaphragm plate **312** as explained below. In other words, the length H**2** of the wavelength plate **314** is length different from the length H**1** of the diaphragm plate **312** in a degree for bending the wavelength plate **314** at the curvature different from the curvature of the diaphragm plate **312**. If the length H**2** of the wavelength plate **314** is set smaller than the length H**1** of the diaphragm plate **312** as in this embodiment, it is possible to set the curvature of the wavelength plate **314** smaller than the curvature of the diaphragm plate **312**.
(36) The sheet holder **363** holds the diaphragm plate **312** such that the center of the opening section **3123** of the diaphragm plate **312** is located on the optical axis of the optical system **31**. The sheet holder **363** holds the diaphragm plate **312** and the wavelength plate **314** in a state in which the diaphragm plate **312** and the wavelength plate **314** are superimposed via a slight gap in the optical axis direction and are bent at the different curvatures. A holding structure for the diaphragm plate **312** and the wavelength plate **314** by the sheet holder **363** is explained in detail below.
(37) The cylindrical lens **313** allows the light transmitted through the wavelength plate **314** and having passed through the opening section **3123** of the diaphragm plate **312** to pass and converges the light in the sub-scanning direction. The second lens holder **362** fixes the cylindrical lens **313** to the housing **36** on the opposite side of the diaphragm plate **312** with respect to the collimator lens **311**. The second lens holder **362** holds the cylindrical lens **313** such that the center of the cylindrical lens **313** is located on the optical axis of the optical system **31**.
(38) As illustrated in FIG. **8**, the sheet holder **363** holding the diaphragm plate **312** and the wavelength plate **314** includes a flat vertical wall section **3631** integrally erected perpendicularly upward from an upper surface **3641** of a bottom wall section **364** of the housing **36**. The vertical wall section **3631** is disposed in parallel to an imaginary surface orthogonal to the optical axis of the optical system **31**.
(39) The vertical wall section **3631** integrally includes a pair of lower end engaging claws **37** and a pair of upper end engaging claws **38** (FIG. **5**) on a surface **3632** on the cylindrical lens **313** side. The lower end engaging claws **37** and the upper end engaging claws **38** only have to be provided in positions deviating from an optical path of light passing through the opening section **3123** of the diaphragm plate **312** and may be provided on the collimator lens **311** side of the vertical wall section **3631**. The lower end engaging claws **37** and the upper end engaging claws **38** are examples of the pair of engaging sections described in the claims of this application. In FIG. **8**, only one of the pair of lower end engaging claws **37** and only one of the pair of upper end engaging claws **38** are illustrated.
(40) The pair of lower end engaging claws **37** is respectively provided to connect the surface **3632**

of the vertical wall section **3631** and the upper surface **3641** of the bottom wall section **364**. The pair of lower end engaging claws **37** is respectively parallel to a YZ plane. The pair of lower end engaging claws **37** is respectively present in positions separated by the same distance in directions separating from each other in the X direction from the center in the X direction of the vertical wall section **3631**. Inner surfaces **371** (only one is illustrated in FIG. **5**) opposed to each other of the pair of lower end engaging claws **37** are examples of the pair of positioning surfaces described in the claims of this application. The distance between the inner surfaces **371** of the pair of lower end engaging claws **37** is slightly larger than the width W**1** of the projecting section **3122** of the diaphragm plate **312** and the width W**2** of the projecting section **3142** of the wavelength plate **314**. The pair of lower end engaging claws **37** has the same shape.

(41) The pair of lower end engaging claws **37** includes engaging grooves **372** extending in the X direction in which the lower end sides **3121** present on both the sides in the X direction of the projecting section **3122** of the diaphragm plate **312** are respectively engaged and the lower end side **3141** present on both the sides in the X direction of the projecting section **3142** of the wavelength plate **314** are engaged. The engaging grooves **372** are present in positions separating in the Y direction from the surface **3632** of the vertical wall section **3631**. The bottoms of the engaging grooves **372** are present in positions separating in the Z direction from the bottom wall section **364**.

(42) If the lower end side **3121** of the diaphragm plate **312** is engaged in the engaging grooves **372** of the pair of lower end engaging claws **37**, the projecting section **3122** of the diaphragm plate **312** is inserted into between the pair of lower end engaging claws **37**. Movement in the X direction of the diaphragm plate **312** can be restricted by the inner surfaces **371** of the pair of lower end engaging claws **37**. If the lower end side **3141** of the wavelength plate **314** is engaged in the engaging grooves **372** of the pair of lower end engaging claws **37**, the projecting section **3142** of the wavelength plate **314** is inserted into between the pair of lower end engaging claws **37**. Movement in the X direction of the wavelength plate **314** can be restricted by the inner surfaces **371** of the pair of lower end engaging claws **37**.

(43) The pair of upper end engaging claws **38** is parallel to the YZ plane and integrally projects in the Y direction from the surface **3632** near the upper end of the vertical wall section **3631**. The pair of upper end engaging claws **38** is respectively present in positions separated the same distance in directions separating from each other in the X direction from the center in the X direction of the vertical wall section **3631**. The distance between the pair of upper end engaging claws **38** is shorter than the distance between the pair of lower end engaging claws **37**. The pair of upper end engaging claws **38** includes engaging grooves **381** extending in the X direction in which an upper end side **3124** of the diaphragm plate **312** is engaged and an upper end side **3144** of the wavelength plate **314** is engaged. The engaging grooves **381** are present in positions separated in the Y direction from the surface **3632** of the vertical wall section **3631**. If the pair of lower end engaging claws **37** is provided on opposite surfaces of the vertical wall section **3631**, the pair of upper end engaging claws **38** only has to be projected from the same surfaces.

(44) The engaging grooves **372** of the lower end engaging claws **37** and the engaging grooves **381** of the upper end engaging claws **38** are generally present in positions opposed in the Z direction. The bottoms of the engaging grooves **372** include, for example, surfaces extended in the horizontal direction and having fixed width in the Y direction. The width in the Y direction of the engaging grooves **372** is at least larger than width obtained by adding up the thickness of the lower end side **3121** of the diaphragm plate **312** and the thickness of the lower end side **3141** of the wavelength plate **314**. The bottoms of the engaging grooves **381** include, for example, surfaces extended in the horizontal direction and having fixed width in the Y direction. The width in the Y direction of the engaging grooves **381** is at least larger than width obtained by adding up the thickness of the upper end side **3124** of the diaphragm plate **312** and the upper end side **3144** of the wavelength plate **314**.

(45) A distance H**3** in the Z direction between the bottoms of the engaging grooves **372** of the lower end engaging claws **37** and the bottoms of the engaging grooves **381** of the upper end

engaging claws **38** is smaller than the length H**1** of the diaphragm plate **312** and smaller than the length H**2** of the wavelength plate **314**. Accordingly, if the lower end side **3121** of the diaphragm plate **312** is engaged in the engaging grooves **372** of the lower end engaging claws **37** and the upper end side **3124** of the diaphragm plate **312** is engaged in the engaging grooves **381** of the upper end engaging claws **38** and the diaphragm plate **312** is attached to the sheet holder **363**, the diaphragm plate **312** bends in an arcuate shape in the direction of the length H**1**.

(46) Similarly, if the lower end side **3141** of the wavelength plate **314** is engaged in the engaging grooves **372** of the lower end engaging claws **37** and the upper end side **3144** of the wavelength plate **314** is engaged in the engaging grooves **381** of the upper end engaging claws **38** and the wavelength plate **314** is attached to the sheet holder **363**, the wavelength plate **314** bends in an arcuate shape in the direction of the length H**2**. In other words, since the diaphragm plate **312** and the wavelength plate **314** are attached to the sheet holder **363** in a bent state, the distance H**3** between the bottoms of the engaging grooves **372** of the lower end engaging claws **37** and the bottoms of the engaging grooves **381** of the upper end engaging claws **38** is set smaller than the length H**1** of the diaphragm plate **312** and smaller than the length H**2** of the wavelength plate **314**.

(47) In this case, the curvature of the diaphragm plate **312** is determined by the length H**1** and the curvature of the wavelength plate **314** is determined by the length H**2**. In this embodiment, H**1**>H**2**>H**3**. As illustrated in FIG. **8**, the diaphragm plate **312** is bent in a direction in which the center in the Z direction of the diaphragm plate **312** swells in the direction of the vertical wall section **3631**, the wavelength plate **314** is bent such that the center in the Z direction of the wavelength plate **314** swells in the same direction as the direction in which the center in the Z direction of the diaphragm plate **312** swells, the wavelength plate **314** is superimposed on the swelling side (the vertical wall section **3631** side) of the diaphragm plate **312**, and the diaphragm plate **312** and the wavelength plate **314** are attached to the sheet holder **363**.

(48) If the diaphragm plate **312** and the wavelength plate **314** are attached to the sheet holder **363** as explained above, the curvature of the wavelength plate **314** is smaller than the curvature of the diaphragm plate **312**, the vicinities of both the ends in the Z direction of the wavelength plate **314** are slightly separated from the diaphragm plate **312**, and a gap is formed between the diaphragm plate **312** and the wavelength plate **314**. The center in the Z direction of the wavelength plate **314** comes into contact with the center in the bending direction of the diaphragm plate **312**.

(49) In this state, the lower end side **3121** of the diaphragm plate **312** is pressed against to engage in the engaging grooves **372** of the lower end engaging claws **37** and the upper end side **3124** of the diaphragm plate **312** is pressed against to engage in the engaging grooves **381** of the upper end engaging claws **38** by a restoration force based on elastic deformation of the diaphragm plate **312**. The center in the Z direction of the wavelength plate **314** is pressed against the diaphragm plate **312**. The wavelength plate **314** adheres to the opening section **3123** and is fixed to the the diaphragm plate **312**.

(50) By attaching the diaphragm plate **312** and the wavelength plate **314** to the sheet holder **363** as explained above, it is possible to fix the diaphragm plate **312** and the wavelength plate **314** to a single sheet holder **363** without using an adhesive, an adhesive tape, or the like. It is possible to reduce the number of components and an assembly manhour and relatively inexpensively assemble the device while maintaining optical performance.

(51) If the diaphragm plate **312** is bent and attached as in this embodiment, it is desirable to set an actual size in the Z direction of the opening section **3123** slightly larger such that a size in the sub-scanning direction of a region in an imaginary XZ plane on which the opening section **3123** is projected in the Y direction in a state in which the diaphragm plate **312** is bent become size for defining a size in the sub-scanning direction of light passing through the opening section **3123**.

(52) If the wavelength plate **314** is bent as in this embodiment, it is considered likely that the optical performance of the wavelength plate **314** slightly changes. Therefore, in this embodiment, the wavelength plate **314** is disposed on the side swelled by the bending of the diaphragm plate **312**

and at least the curvature of the wavelength plate **314** is set smaller than the curvature of the diaphragm plate **312** to prevent the change in the optical performance of the wavelength plate **314** due to the bending.

(53) The wavelength plate **314** is disposed to be superimposed on the swelling side of the diaphragm plate **312** to prevent a deficiency in which the projecting section **3142** of the wavelength plate **314** and the projecting section **3122** of the diaphragm plate **312** interfere. If the curvatures of the wavelength plate **314** and the diaphragm plate **312** are maintained and superimposition order of the wavelength plate **314** and the diaphragm plate **312** is reversed, the projecting section **3142** and the projecting section **3122** interfere and undesired stress acts on the wavelength plate **314** having lower rigidity (easily bent) than the diaphragm plate **312**. Therefore, in this embodiment, the wavelength plate **314** and the diaphragm plate **312** are superimposed in superimposition order for preventing the projecting section **3142** and the projecting section **3122** from interfering.

(54) In the embodiment explained above, a case is explained in which the diaphragm plate **312** and the wavelength plate **314** are attached to the sheet holder **363** at the curvatures and in the superimposition order illustrated in FIG. **8**. However, the diaphragm plate **312** and the wavelength plate **314** can be attached to the sheet holder **363** even if the superimposition order, the curvatures, the bending direction, and the like are changed. Modifications of the method of attaching the diaphragm plate **312** and the wavelength plate **314** are explained below.

(55) In a first modification, the shape of the wavelength plate **314** is changed to a shape illustrated in FIG. **9**. A wavelength plate **40** illustrated in FIG. **9** includes substantially rectangular two projecting sections **42** projecting in the Z direction from both the ends in the X direction of a lower end side **41** of the wavelength plate **40**. Since the two projecting sections **42** are disposed on the outer side in the X direction of the pair of lower end engaging claws **37** of the sheet holder **363**, the wavelength plate **40** has width W**3** in the X direction larger than the width of the wavelength plate **314**. A distance W**4** between the two projecting sections **42** is slightly longer than the distance between the outer surfaces in the X direction of the pair of lower end engaging claws **37**. In order to set a curvature of the wavelength plate **40** larger than the curvature of the diaphragm plate **312**, length H**4** in the Z direction excluding the two projecting sections **42** is set slightly larger than the length H**1** of the diaphragm plate **312**.

(56) In the first modification, as illustrated in FIG. **10**, the wavelength plate **40** is attached to the sheet holder **363**. The diaphragm plate **312** is attached in the same manner as in the embodiment explained above. Superimposition order of the wavelength plate **40** and the diaphragm plate **312** is also the same as the superimposition order in the embodiment explained above. The wavelength plate **40** is disposed to be superimposed on the swelling side of the diaphragm plate **312**. Since the length H**4** of the wavelength plate **40** is set larger than the length H**1** of the diaphragm plate **312**, the curvature of the wavelength plate **40** is larger than the curvature of the diaphragm plate **312**. A gap is formed between the diaphragm plate **312** and the wavelength plate **40** in the center in the Z direction. Since the two projecting sections **42** of the wavelength plate **40** are disposed on the outer side of the pair of lower end engaging claws **37**, the two projecting sections **42** of the wavelength plate **40** and the projecting section **3122** of the diaphragm plate **312** do not interfere.

(57) According to the first modification, the curvature of the wavelength plate **40** is slightly larger than the curvature of the wavelength plate **314** in the embodiment explained above. However, the wavelength plate **40** can be attached to the single sheet holder **363** together with the diaphragm plate **312**. It is possible to achieve the same effects as the effects in the embodiment explained above. According to the first modification, since the curvature of the wavelength plate **40** is increased, the restoration force based on the elastic deformation can be increased. An engaging force of the wavelength plate **40** with the lower end engaging claws **37** and the upper end engaging claws **38** can be increased. The wavelength plate **40** can be firmly fixed by the sheet holder **363**.

(58) In a second modification, the shape of the diaphragm plate **312** is changed to a shape illustrated in FIG. **11** and the shape of the wavelength plate **314** is changed to a shape illustrated in

FIG. **12**. A diaphragm plate **50** illustrated in FIG. **11** has the same shape as the diaphragm plate **312** in the embodiment explained above except that the diaphragm plate **50** includes substantially rectangular two projecting sections **51** projecting from the lower end side **3121** of the diaphragm plate **50**. The two projecting sections **51** are formed by cutting out the center in the X direction of the projecting section **3122** of the diaphragm plate **312** at width W**5**.
(59) A wavelength plate **60** illustrated in FIG. **12** includes a substantially rectangular projecting section **62** projecting from the center in the X direction of a lower end side **61** of the wavelength plate **60**. Width W**6** of the projecting section **62** is slightly smaller than the width W**5** of the cutout portion of the diaphragm plate **50**. The length in the Z direction of the wavelength plate **60** is H**4** and is the same as the length of the wavelength plate **40** in the first modification.
(60) In the second modification, as illustrated in FIG. **13**, the diaphragm plate **50** and the wavelength plate **60** are attached to the sheet holder **363**. The diaphragm plate **50** is attached in the same manner as the diaphragm plate **312** explained above. Superimposition order of the diagraph plate **50** and the wavelength plate **60** is the same as the superimposition order in the first modification. The wavelength plate **60** is disposed to be superimposed on a swelling side of the diaphragm plate **50**. In the second modification, the projecting section **62** of the wavelength plate **60** is inserted into between the two projecting section **51** of the diaphragm plate **50**. Consequently, as in the first modification, it is possible to prevent a deficiency in which the projecting section **51** of the diaphragm plate **50** and the projecting section **62** of the wavelength plate **60** interfere.
(61) In the embodiment and the modifications thereof explained above, bending directions of the diaphragm plate **312** (**50**) and the wavelength plate **314** (**40**, **60**) may be reversed. From the viewpoint of assembly work, the bending directions of the diaphragm plate **312** (**50**) and the wavelength plate **314** (**40**, **60**) are desirably set in the directions in the embodiment and the modifications explained above.
(62) In the embodiment and the modifications thereof explained above, the lengths and the curvatures of the diaphragm plate **312** (**50**) and the wavelength plate **314** (**40**, **60**) are differentiated. However, a diaphragm plate and a wavelength plate may be bent at the same curvature and attached to be superimposed without a gap.
(63) Further, in the embodiment and the modifications thereof, a polarizing plate may be incorporated in the device instead of the wavelength plate **314** (**40**, **60**).
(64) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of invention. Indeed, the novel apparatus and methods described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the apparatus and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. An optical scanning device, comprising: a light source configured to emit light; a first optical sheet disposed on an optical path of the light emitted from the light source in a state in which the first optical sheet is bent by elastic deformation; a second optical sheet disposed on the optical path, configured to transmit the light, and bendable by elastic deformation; a holding section including a pair of engaging members with which both end portions in a bending direction of the first optical sheet are engaged by a restoration force based on the elastic deformation of the first optical sheet and with which both end portions in a bending direction of the second optical sheet are engaged by a restoration force based on the elastic deformation of the second optical sheet in a state in which the second optical sheet is bent at a curvature different from a curvature of the first optical sheet and in a same direction as the bending direction of the first optical sheet, the holding section

holding the first optical sheet and the second optical sheet in a state in which the first optical sheet and the second optical sheet are superimposed via a gap; and a polygon mirror configured to reflect the light from the light source guided via the first optical sheet and the second optical sheet and rotate to scan a scan surface present in a reflecting direction of the light.

2. The optical scanning device according to claim 1, wherein the light source is a laser array including a plurality of light emitting elements disposed side by side in a scanning direction by the polygon mirror.

3. The optical scanning device according to claim 1, wherein the first optical sheet is a diaphragm plate including an opening section that allows the light to pass, and the second optical sheet is a wavelength plate that transmits the light and changes a polarization direction of the light.

4. The optical scanning device according to claim 3, wherein the wavelength plate is bent at a curvature smaller than a curvature of the diaphragm plate and is superimposed on a side swelled by the bending of the diaphragm plate.

5. The optical scanning device according to claim 4, wherein a center in the bending direction of the wavelength plate is in contact with a center in the bending direction of the diaphragm plate.

6. The optical scanning device according to claim 4, wherein the diagraph plate includes a first projecting section projecting from one end in the bending direction to a first direction orthogonal to the one end, the wavelength plate includes a second projecting section projecting from one end in the bending direction opposed to the one end of the diaphragm plate to the first direction orthogonal to the one end, one of the engaging members, with which the one end of the diaphragm plate and the one end of the wavelength plate engage, of the holding section includes a pair of positioning surfaces that locks both ends in a second direction orthogonal to the first direction of the first projecting section and restricts movement of the first projecting section in the second direction, and the pair of positioning surfaces locks both ends in the second direction of the second projecting section of the wavelength plate and restricts movement of the second projecting section in the second direction.

7. The optical scanning device according to claim 6, wherein a center in the bending direction of the wavelength plate is in contact with a center in the bending direction of the diaphragm plate.

8. The optical scanning device according to claim 3, wherein the wavelength plate is a half wavelength plate that changes the polarization direction of the light by 45°.

9. The optical scanning device according to claim 3, wherein the wavelength plate is a quarter wavelength plate that changes linearly polarized light to circularly polarized light.

10. The optical scanning device according to claim 3, wherein the wavelength plate is a wavelength plate obtained by sandwiching a double refraction film with polyethylene terephthalate sheets.

11. An image forming apparatus, comprising: a document feeder; a photoconductive drum; a developing device; a fixing device; an optical scanning device, comprising: a light source configured to emit light; a first optical sheet disposed on an optical path of the light emitted from the light source in a state in which the first optical sheet is bent by elastic deformation; a second optical sheet disposed on the optical path, configured to transmit the light, and bendable by elastic deformation; a holding section including a pair of engaging members with which both end portions in a bending direction of the first optical sheet are engaged by a restoration force based on the elastic deformation of the first optical sheet and with which both end portions in a bending direction of the second optical sheet are engaged by a restoration force based on the elastic deformation of the second optical sheet in a state in which the second optical sheet is bent at a curvature different from a curvature of the first optical sheet and in a same direction as the bending direction of the first optical sheet, the holding section holding the first optical sheet and the second optical sheet in a state in which the first optical sheet and the second optical sheet are superimposed via a gap; and a polygon mirror configured to reflect the light from the light source guided via the first optical sheet and the second optical sheet and rotate to scan a scan surface present in a reflecting direction of the light; and a paper discharge tray.

12. The image forming apparatus according to claim 11, wherein the light source is a laser array including a plurality of light emitting elements disposed side by side in a scanning direction by the polygon mirror.
13. The image forming apparatus according to claim 11, wherein the first optical sheet is a diaphragm plate including an opening section that allows the light to pass, and the second optical sheet is a wavelength plate that transmits the light and changes a polarization direction of the light.
14. The image forming apparatus according to claim 13, wherein the wavelength plate is bent at a curvature smaller than a curvature of the diaphragm plate and is superimposed on a side swelled by the bending of the diaphragm plate.
15. The image forming apparatus according to claim 14, wherein a center in the bending direction of the wavelength plate is in contact with a center in the bending direction of the diaphragm plate.
16. The image forming apparatus according to claim 14, wherein the diagraph plate includes a first projecting section projecting from one end in the bending direction to a first direction orthogonal to the one end, the wavelength plate includes a second projecting section projecting from one end in the bending direction opposed to the one end of the diaphragm plate to the first direction orthogonal to the one end, one of the engaging members, with which the one end of the diaphragm plate and the one end of the wavelength plate engage, of the holding section includes a pair of positioning surfaces that locks both ends in a second direction orthogonal to the first direction of the first projecting section and restricts movement of the first projecting section in the second direction, and the pair of positioning surfaces locks both ends in the second direction of the second projecting section of the wavelength plate and restricts movement of the second projecting section in the second direction.
17. The image forming apparatus according to claim 16, wherein a center in the bending direction of the wavelength plate is in contact with a center in the bending direction of the diaphragm plate.
18. The image forming apparatus according to claim 13, wherein the wavelength plate is a half wavelength plate that changes the polarization direction of the light by 45°.
19. The image forming apparatus according to claim 13, wherein the wavelength plate is a quarter wavelength plate that changes linearly polarized light to circularly polarized light.
20. The image forming apparatus according to claim 13, wherein the wavelength plate is a wavelength plate obtained by sandwiching a double refraction film with polyethylene terephthalate sheets.